Placement Is Provenance: Anchor-Preserving Ingestion for Figure-Grounded Retrieval-Augmented Generation

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Abstract—Multimodal retrieval-augmented generation is evaluated almost entirely on which evidence is retrieved. A tutoring system that shows figures from a textbook is judged on a second axis as well: where the figure lands in the explanation. We show that this axis is separable, that it is cheap to measure, and that the information determining it is produced by every layout parser and routinely discarded before serving.

The method is a join over data the parser already produced, not a model: it is independent of language and of document genre. We evaluate it on two public benchmarks with a configuration frozen before either dataset was obtained. On MRAMG-Bench's Academic domain a 568M-parameter cross-encoder over anchor-derived context reaches Image Precision 69.12 against 65.28 for the best published system (GPT-4o) — higher, though a question-level bootstrap shows the four-point margin is not statistically resolvable at $n = 200$ — and diluting the anchor window degrades precision monotonically, which is the mechanism claim in falsifiable form. On MMDocRAG, a vision-language gate run against gold we did not author adds +13.65 precision over the identical pipeline without it. Under the benchmark's own end-to-end generation protocol, blind LLM judging finds that a mechanical anchor join places figures exactly as well as the generator's own inline choice (48 questions to 49) while remaining traceable to a source position: placement does not need the generator. A rule study over 6,819 questions further shows that absolute admission thresholds do not transfer across corpora while any scale-free rule does — worth sixteen macro-F1 points, and replicated out of sample after pre-registration. Losses are reported in the same tables as wins, including a domain the method loses and an end-to-end F1 comparison it loses.

For development and stress testing we release a caption-scarce stress corpus of Indonesian K–12 curriculum textbooks: 13 government titles, 2,816 pages, extracted through two independent pipelines, with 486 adversarially filtered questions. This corpus is not the scope of the claim but its hardest case: only a fifth of its figures carry a printed caption, so the failure this paper is about is visible there and largely hidden in caption-rich corpora. We introduce a layout-gold placement metric that derives placement ground truth from the source document's own typesetting at no per-item annotation cost, and a headline Grounded Figure F1 that credits a figure only when it is both the right figure and in the right place.

Two findings require no model and replicate across both extraction pipelines. Only 19.3 %/34.3 % of figures carry a printed caption, while 100 % carry a recoverable reading-order anchor. And under perfect retrieval and a perfect answer, caption-based placement is still off by 2.08/4.69 sentences. Placement is not implied by retrieval. Notably, the corpus with nearly twice the caption coverage places worse, because its share of lexically indistinguishable figures also doubles: caption availability is not caption discriminability.

On the textbook corpus, composing a vision-language filter with that same cross-encoder raises figure-selection precision from 0.028 to 0.542 against a preliminary human annotation ($n = 48$) — sight for recall, discrimination for precision — a result the MMDocRAG gate run now corroborates externally.

Index Terms—Multimodal retrieval-augmented generation, document layout analysis, figure grounding, document ingestion, cross-encoder reranking, admission calibration.

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I. Introduction

An AI study assistant grounded in school textbooks must do more than quote the right paragraph. Curriculum books teach substantially through pictures: a labelled water-cycle diagram, a food-chain chart, a worked geometric construction. When such a system answers a child's question, showing the right figure at the right point in the explanation is part of the answer, not decoration around it.

Two distinct things can go wrong. The system can show the wrong figure, or it can show the right figure in the wrong place — after the explanation has moved on, or bundled at the end where the reader must reconstruct which sentence it belongs to. The first failure is measured everywhere. The second is measured almost nowhere.

This paper makes five claims, each falsifiable by the experiments reported here.

- C1 Figures in curriculum textbooks carry content absent from the prose, so text-only retrieval has a ceiling that better text retrieval cannot raise.
- C2 Placement is a failure mode distinct from retrieval. It can be made to fail while retrieval and generation are both perfect.
- C3 The signal that fixes it — the figure's position in the parser's reading-order block sequence — is produced by every layout parser, is recoverable for 100 % of figures, and is discarded by conventional ingestion.
- C4 Selection, not placement, is where the practical gain lies, and the gain requires a stage that looks at the image rather than at text about the image.
- C5 Admission — how many ranked candidates to emit — does not transfer across corpora when ruled by an absolute score threshold, and does under any scale-free rule. The choice of rule family is worth an order of magnitude more than any tuning within a family.

C2 and C3 are established model-free. C1, C4 and C5 are tested on two public benchmarks — MRAMG-Bench and MMDocRAG, spanning seven evaluation domains — under a configuration frozen before either dataset was obtained, including an end-to-end run of MRAMG's own generation protocol with blind LLM judging. The same experiments produce the comparisons the method loses, and Section XII lists them beside the wins.

A. Setting

The method is corpus-agnostic; the deployment that motivated it is not. It is an on-premise study assistant for an Indonesian university partner, serving national curriculum textbooks on an air-gapped NVIDIA GB10 host. Every model in the serving path is 8B-class or smaller and runs locally. This constrains the design — no frontier model is available at answer time — and makes the efficiency results in Section VIII load-bearing rather than incidental.

II. Related Work

A. Multimodal RAG benchmarks

Recent benchmarks evaluate document-centric multimodal RAG along retrieval and answer quality. UniDocBench [1] unifies document-centric evaluation; REAL-MM-RAG [3] targets retrieval robustness under query rephrasing; ColPali [6] introduces late-interaction visual retrieval and the ViDoRe benchmark. MIRACL-VISION [10] extends visual retrieval to 18 languages and reports substantial degradation outside English.

These evaluate which evidence is found. None scores where a figure is placed within a generated answer, which is the axis this paper isolates.

B. Figure placement in generated answers

MRAMG-Bench [7] is the closest prior work: it scores inserted images and supplies a bipartite sentence–image assignment rule, which we implement as a comparator. MMDocRAG [2] reports image-quote selection F1 separately from text. VinQA [9] defines a visual citation measure with a citation-order placement rule, implemented here as a second comparator. ImageRef-VL [5] addresses contextual image referencing in vision-language models. RAG-IGBench [8] evaluates open-domain interleaved image–text generation with automated metrics for text, image and their consistency, validated against human assessment; like MRAMG's judged metrics, its placement signal ultimately rests on a model or a person reading the answer, whereas the layout-gold metric here derives it from the source typesetting.

Our contribution relative to these is (i) a placement ground truth obtained from the document's own typesetting rather than from annotation, and (ii) the observation that the ingestion step, not the ranking step, is where the necessary signal is lost.

C. Caption-scarce document regimes

The benchmarks above are caption-rich: their figures overwhelmingly carry printed captions, and a caption-matching pipeline is therefore never tested where it actually breaks. We are not aware of a published multimodal RAG evaluation in which most figures have no caption at all. The corpus released here is selected for that property — 19.3% caption coverage, alongside dense decorative imagery — and not for its language, which is incidental to every claim we make.

III. Method

A. The anchor, and where it is lost

A layout parser emits, for each page, an ordered sequence of blocks: headings, paragraphs, captions, figures. A figure's position in that sequence is a precise statement of where in the reading flow it belongs. Conventional ingestion then stores the prose as chunks and the figures as a separate array, and the ordering relation between them is dropped.

Fig. 1 shows the loss and the alternative side by side. We define the anchor of a figure as the index of the text chunk containing the prose block that immediately precedes it in reading order. Anchoring requires no model: it is a join over data the parser already produced. We recover it by matching a word-prefix of the preceding block against chunk text, shortening the probe from ten words to four until a match is found.

Anchors are preserved through ingestion into chunk metadata, and consumed at retrieval time: a figure whose anchor chunk was retrieved is admitted as a candidate before any caption test is applied. This ordering matters, because a caption test cannot admit the 66–81 % of figures that carry no printed caption.

B. The pipeline, stage by stage

The method has two halves, summarised in Fig. 2. Ingestion runs once per document and is model-free. Serving runs per query and adds two ranking stages. Inputs and outputs are stated for each stage so that any of them can be replaced independently.

Ingestion (once per document).

1) Parse. A layout parser returns, per page, an ordered sequence of typed blocks $B = \langle b_1, \ldots, b_n \rangle$ with $b_i \in \{\text{heading}, \text{paragraph}, \text{caption}, \text{figure}, \text{table}\}$. This ordering is the only input the method needs and every layout parser produces it.
2) Chunk. Prose blocks are segmented into retrieval chunks $C = \langle c_1, \ldots, c_m \rangle$, each recording the block indices it spans.
3) Anchor. For each figure block b_f, let b_p be the nearest preceding non-figure prose block. The anchor $a(f)$ is the index of the chunk containing b_p. Resolution is by word-prefix match: the first ten words of b_p are matched against chunk text, shortening the probe to four words until a unique match is found. Cost is one pass over blocks; no model is invoked.

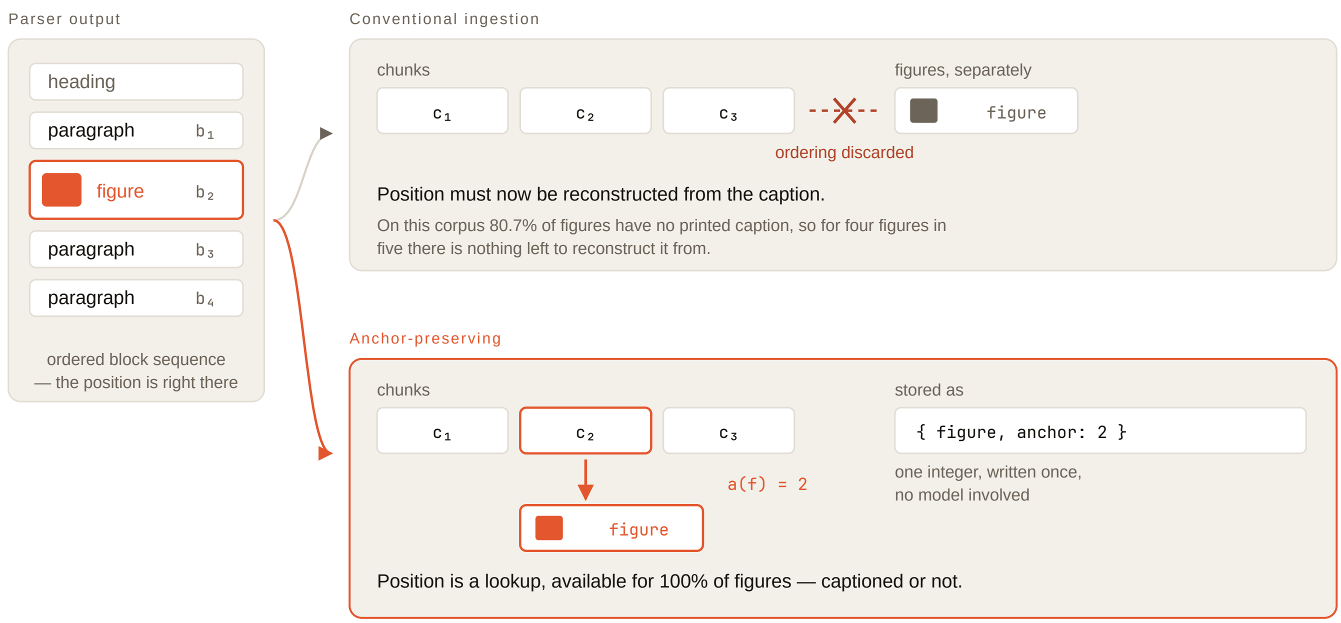

Fig. 1. Where the reading order is lost. The parser emits an ordered block sequence in which the figure's position is already stated. Conventional ingestion stores prose as chunks and figures as a separate array and drops the relation between them, so placement must afterwards be reconstructed from a caption that 80.7 % of figures in this corpus do not have. Anchor-preserving ingestion records the index of the chunk holding the preceding prose block — one integer, written once, with no model invoked — and placement becomes a lookup available for every figure.

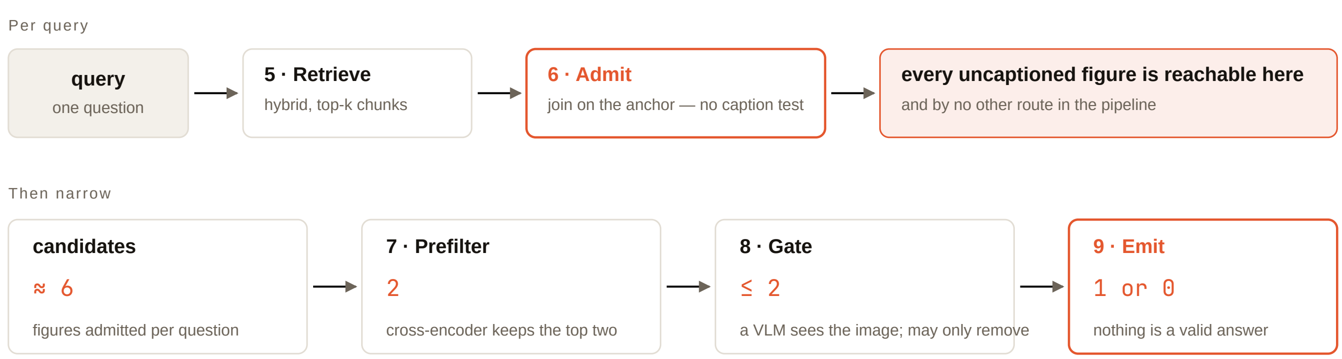

Fig. 2. Serving. Admission (stage 6) is a join on the anchor rather than a caption test, and is the only route by which a candidate enters; every later stage can only narrow. The cross-encoder truncates to two candidates, the vision-language gate may remove but never add or reorder, and emitting nothing is a valid outcome.

4) Persist. $a(f)$ is written to the figure record and mirrored into the chunk's metadata, so retrieval of $c_{a(f)}$ recovers f by lookup rather than by search.

Serving (per query).

5) Retrieve. Hybrid retrieval returns the top-k chunks for query q.
6) Admit. Every figure whose anchor chunk is in that set becomes a candidate. Admission is a join on $a(f)$, so it is independent of whether f has a caption — which is what allows the 66–81 % of uncaptioned figures to be reachable at all. No candidate is admitted by any other route.
7) Prefilter. A cross-encoder scores $(q, t(f))$ for each candidate, where $t(f)$ is the figure's index text. Candidates are ranked and truncated to the top N. We use $N = 2$; Section VIII shows this matches the full candidate set at a third of the latency.
8) Gate. Each surviving candidate is shown as an image to a vision-language model with a strict yes/no prompt: does this figure help answer q? Rejected candidates are dropped. The gate can only remove, never add or reorder, so its worst case is the prefilter's output.
9) Rank and emit. Survivors are ordered by their stage-7 score and the top one is emitted. Emitting nothing is a valid and frequent outcome.
10) Place. The emitted figure is inserted at the sentence boundary corresponding to $a(f)$ in the generated

answer, rather than at a position inferred from its caption.

Stages 3 and 6 are the contribution. Stages 7–9 are a composition of existing components, and Section VIII reports what each contributes.

C. Layout-gold: the document annotates itself

Placement ground truth normally requires a human to say where a figure belongs in a generated answer. We avoid this. Take a figure's own anchor chunk as the answer text. Retrieval is then perfect by construction, the answer is perfect, and the correct insertion slot is known exactly from the block sequence. Any residual error is intrinsic to the placement mechanism, because both the retriever and the generator have been removed.

This yields placement ground truth for every figure in the corpus at zero per-item annotation cost, and it is what makes the model-free results of Section V possible.

D. Metrics

Placement displacement (PD) is the signed distance, in sentences, between where a figure was placed and its layout-gold slot; we report |PD| and the exact-placement rate. Selection precision/recall/F1 score the figure set. Grounded Figure F1 credits a figure only when it is both selected correctly and placed within tolerance.

E. LLM-as-judge, and its limits

Human annotation at scale was not affordable, so a vision-language judge was calibrated against the human annotation that was. We report Cohen's κ against a human annotator before using any judge output. Our model-generated harness gold agrees with a human at $\kappa = 0.092$ — chance. A frontier VLM judge reaches $\kappa = 0.552$ on the eight-candidate annotation protocol and $\kappa = 0.662$ on the six-candidate one, the difference being consistent with the smaller pool posing an easier discrimination; permissiveness is 1.38× against 2.9× for an on-premise 8B VLM on the same task. Every scaled number carries the κ of the instrument that produced it.

The credential is per-model, and we can now show it does not transfer across vendors either: a second frontier model (MiniMax-M3) sat the identical exam — same protocol, same pools, same images, 360 human-labelled pairs — and scored $\kappa = 0.362$ at 2.76× permissiveness, against 0.556 at 1.24× for the incumbent. It was therefore not licensed as a selection judge or gate in this study. A κ measured for one model licenses that model alone; swapping the judge silently swaps the instrument.

IV. Experimental Setup

A. Corpus

Thirteen Indonesian government curriculum titles, 2,816 pages, spanning grades 1, 4 and 8. Each book was extracted through two independent pipelines: a hosted OCR service and an on-premise MinerU sidecar. These use different models, different figure detectors and different noise filtering, and neither corpus was derived from the other. Agreement between them is therefore replication, not re-reporting.

TABLE I
Model-free corpus measures, computed independently on two extraction pipelines. Agreement is replication, not re-reporting.

	Measure	Hosted	On-prem
—	Usable figures	1,552	1,226
A1	With a printed caption	19.3 %	34.3 %
A2	Lexically indistinguishable	20.7 %	34.1 %
A3	Shares a page with another figure	49.9 %	54.2 %
A3b	Anchor vs. page resolution	1.49×	1.56×
—	With a reading-order anchor	100 %	100 %
A4	Caption selection F1, oracle retrieval	0.254	0.232
A5	Caption |PD|, oracle answer	2.08	4.69
A5b	End-of-answer |PD|	8.71	12.13

B. What is released

We release the harness, the metric implementations, the question sets and the derived per-figure measurements. We do not redistribute the textbooks: they are Indonesian government publications carrying their own terms, and every structural measure in Section V is reproducible from the cached extraction we do release without re-obtaining them.

C. Questions

486 questions, adversarially filtered to remove items answerable from the question alone, from front matter, or from pedagogical apparatus. A 48-item subset carries human figure annotation (19 positive figure–question links); this subset is the only non-circular gold available and is used for every headline selection number.

D. Systems

Ten configurations, including implementations of the MRAMG bipartite placement rule and the VinQA citation-order rule. Retrieval is hybrid dense/sparse with bge-m3 embeddings at dimension 1024; reranking uses bge-reranker-v2-m3 (568M parameters) served locally. Generation uses an 8B-class SEA-LION vision-language model.

V. Structural Results

These come from the corpus alone: deterministic, reproducible offline, and impossible to tune. Table I reports both extraction paths.

Four consequences follow without running any system.

The caption signal is mostly absent (A1). Four figures in five carry no printed caption on the hosted path, so a caption-based mechanism must fall back on page prose — which is shared by every figure on that page.

A fifth of the corpus is unsolvable by caption matching in principle (A2). For 20.7 % of figures, another figure in the same book has an effectively identical index text. No matcher can distinguish them. This is an upper bound on a competing approach, derived without running it.

Page-level provenance cannot resolve half the corpus (A3). Half of all figures share a page with another, so page-level attribution is ambiguous for 49.9 % of cases. Anchoring provides 1.49× the positional resolution, and that gain is concentrated exactly where page-level methods fail.

Placement fails when nothing else does (A5) — this is C2, established without a model. Under oracle retrieval and an oracle answer, caption matching still lands 2.08 sentences away and is exactly right only 43.0 % of the time; a design with no positional signal at all sits 8.71 sentences away and is never exactly right.

More captions did not help. The on-premise path has nearly twice the caption coverage (34.3 % vs. 19.3 %) yet places worse (4.69 vs. 2.08 sentences). The reason is visible in A2: its share of lexically indistinguishable figures also nearly doubles. Caption availability is not caption discriminability, and only the latter could help. This would have been invisible on a single corpus.

Fairness note. A first implementation of A4 scored the caption mechanism at F1 0.025 by taking the first three keyword matches in document order. Ranking by keyword-overlap count instead — the charitable reading — raised it tenfold to 0.254. The reported figure is the charitable one.

Tautology note. Under both oracle setups the anchor mechanism scores perfectly by construction, because its selection rule and the gold set coincide. That number is not evidence and is stated only to make the circularity explicit.

VI. External Validation

Every result above is measured on a corpus we assembled against a gold standard we defined. 80 % of our gold figure links are derived from the anchor itself, and on those links our harness gold agrees with a human annotator at chance ($\kappa = 0.092$). Internal care cannot repair that; only measurement on data we did not author can.

MRAMG-Bench [7] provides it, and its Academic subset is the strongest form of it available to us: those gold image lists were constructed by ten graduate-student annotators in computer science, each assigned papers within their own area, and passed through a three-stage human verification. Where our harness gold agrees with a person at chance, this reference is what a panel of domain experts chose. Candidates are the images of the question's provenance document, each represented by the prose immediately preceding its placeholder — the same anchor our method uses, and the same information available to their placeholder-based baseline. The configuration was fixed on our own data before this dataset was obtained; no parameter was tuned on MRAMG. Table II reports the result against the comparators published with that benchmark.

The margin, bounded. A bootstrap over questions ($B = 2 \times 10^4$) places our pooled Academic precision at 69.12 with 95 % interval $[61.43, 76.56]$; 15.7 % of resamples fall at or below the published 65.28. The comparator is a point estimate whose own sampling error we cannot resample, so the test is one-sided by necessity. The honest verdict is higher, not significantly higher: on 200 questions a four-point margin is real but not resolvable, and we decline to claim more than the interval supports.

TABLE II
MRAMG-Bench Academic subset, Image Precision ($n = 200$). Published figures for comparators.

System	Image Precision
Ours (cross-encoder, 568M params)	69.12
Ours, forced emission (0 % silent)	67.50
GPT-4o, LLM-based	65.28
Claude-3.5-Sonnet, LLM-based	62.17
GPT-4o, MLLM-based	60.39
Gemini-1.5-Pro, LLM-based	59.85
DeepSeek-V3, rule-based	56.12
Llama-3.3-70B (best open-weight)	38.78
Qwen2-VL-7B, MLLM-based	1.63

The abstention objection, and its test. Image Precision counts only emitted images, so a system that declines on hard questions could purchase its score; ours abstains on 49 %. Removing that advantage entirely — emitting exactly one image on every question — yields precision 67.50, recall 61.64, F1 64.44. Still ahead of every published configuration, with recall rising from 41.59. The margin is not an artefact of selective answering.

The other five subsets. One subset reported out of six reads as selection whether or not it was, so Table III reports all of them, pooled into the three domains the published comparison uses. The pooling sums raw counts — averaging per-subset precision would weight a 200-question subset like a 2360-question one. One domain is a win, one is a loss, and one admits no claim at all: on every Web-domain question, every candidate our construction produces is already a gold image (the all-gold column), so there is no wrong answer available and precision is arithmetic rather than earned. Quoting our 100.00 there against the published 93.63 would be a false claim, and the diagnostic exists so that it cannot be made silently. The Lifestyle loss is informative in an unexpected direction: the manual subset carries 91.5 candidates per question — 27× the arxiv pool — and still scores higher than recipe at 4.9 candidates, so distractor count is not what breaks the method. Recipe averages 2.2 gold images per question while the frozen configuration emits at most two and abstains on 19 %, capping recall; Section VII locates most of the remaining loss in the admission rule rather than the ranking.

Scope. These are selection results placed beside end-to-end systems that also generate the surrounding answer; Image Precision is computed over the same quantity, but the tasks are not identical. (69.12 is the number this paper quotes throughout: the CPU reproduction, whose per-question data the bootstrap uses. The original GPU run scored 69.63 — identical true positives, recall and silent rate, differing by exactly one borderline emission —

TABLE III
All MRAMG-Bench domains under the frozen configuration. All-gold is the share of questions where every candidate is already a gold image; where it is 100 % the domain poses no selection decision under our candidate construction, and no claim is made.

Domain	n	All-gold	Ours IP	Best publ.	
Academic	200	0 %	69.12	65.28	win
Lifestyle	2714	41 %	55.00	62.23	loss
Web	1850	100 %	(100.00)	93.63	no claim

and is kept in the released artifacts.)

The mechanism, confirmed by making it worse. If the anchor carries the signal, diluting it should destroy it. It does, monotonically: representing each image by a growing window of prose around its position, Academic Image Precision decays 72.67 (200 characters) → 69.12 (400) → 59.59 (800) → 58.74 (1600) → 57.30 (6000, effectively the whole document). Below the peak the failure flips from dilution to starvation — at 100 characters precision holds (70.97) but recall collapses from 51.09 to 19.82, as windows shrink beneath the admission floor. The local prose adjacent to the figure's position is what discriminates; every character beyond it is noise, and too few characters leave nothing to score. Two hygiene notes. The frozen configuration uses 400, and the sweep shows 200 would have scored higher on both precision and recall — we do not adopt it, because a value discovered on the test set would convert every number in this section from measurement to tuning. And the shape is itself the strongest form of the claim: had the anchor been incidental, some wider window would have won.

What it does not license. The same configuration reaches ≈30 % precision on the Indonesian textbooks. That corpus is harder: a third of its figures carry no caption, most are decorative, and a dozen illustrations per book depict children learning. A benchmark number must not be quoted as a product capability.

A. MMDocRAG: a second benchmark, and what the ranker is not

A single external benchmark is a single point of failure, so we add MMDocRAG [2]: 2,055 questions over 147 documents in four domains (financial, academic, news, research reports), 5.0 image candidates per question against 1.5 gold, expert-annotated by its authors, and only 0.5 % of questions degenerate — the selection decision is real almost everywhere. One property makes it the ideal control: each candidate ships with an author-written description, so the text standing in for an image is theirs, not ours, and the regime is the opposite of the textbook corpus — text about the image is present and good.

In that regime the cross-encoder earns nothing over lexical matching: 63.38 F1 against 63.48 for BM25 over the same descriptions (top-2, no threshold, whole benchmark). This is a result we want on record, because it disciplines the Academic win: what carried MRAMG was not the ranker but the representation it ranked — anchor-derived context standing in for images that captions do not describe. Where good text about the image already exists, a 568M cross-encoder is BM25.

The frozen configuration also exposes its own admission floor here. With the absolute threshold ($s \geq 0.1$) it scores 57.30 overall and 11.31 on the News domain; removing only the floor raises News to 74.03. A constant admissible-score cut assumes the score distribution is corpus-invariant, and it is not — Section VII treats this as a finding about admission rules rather than a number to be quietly fixed.

B. The vision gate, on gold we did not author

Every external number above tests the weakest configuration we own: the cross-encoder alone. The stage that carried the internal result — a vision-language model that sees each surviving candidate as an image and may only remove it — had never been run against external gold. MMDocRAG is the hard case for it: descriptions here are good, and the gate's value was measured where they are absent. If sight helps anyway, the claim is stronger than we made it.

On a 200-question stride sample (the file is document-ordered, so a prefix would sample a handful of documents rather than the benchmark), with an identical prefilter and top-1 emission for both arms:

System	IP	IR	IF1
Cross-encoder alone	75.00	51.55	61.10
+ vision gate (composed)	88.65	56.36	68.91

The gate saw 400 candidates and kept 228. Precision rising is expected of a remove-only stage; recall rising (+4.8) is not, and the mechanism is worth stating: under top-1 emission, deleting a wrongly ranked leader promotes the next survivor into the emitted slot. We measured eight such promotions against one loss. Removal never reorders the list, but it reorders what is emitted.

Two honesty notes. The gate instance here is a frontier VLM, not the 8B-class model of the deployed system — what is validated externally is the architecture (sight composed after discrimination), not the deployed checkpoint. And the composed 68.91 remains behind the 80.7 the benchmark's authors report for GPT-4.1 selecting over interleaved candidates; the comparison we claim is against our own ablation on identical inputs, where the gate is worth +13.65 precision and +7.81 F1.

C. End-to-end, and where the pipeline does not win

Every external number above is a selection result placed beside end-to-end systems, and we have said so each time. This subsection removes the caveat by running MRAMG's own generation protocol on Academic: bge-m3 retrieval over ∼256-token sentence chunks, top-10 chunks per question, an 8B SEA-LION generator on the partner-class GPU, and two arms differing in exactly one mechanism. In arm A — the benchmark's LLM-based framework — the retrieved chunks keep inline image placeholders and the generator inserts them while writing, per the benchmark's published prompt. In arm B the same chunks are stripped of placeholders, the same generator writes a text-only answer, and the frozen IKAT pipeline selects images by anchor context. Same retriever, same chunks, same generator, same questions; retrieval surfaces 90.4 % of gold images, which ceilings both arms equally.

Arm	IP	IR	IF1	R-L	silent
A: generator inserts	53.38	73.13	61.71	0.261	7 %
B: IKAT, frozen	63.91	38.81	48.30	0.277	50 %
B: IKAT, scale-free adm.	47.45	68.04	55.91	0.277	0 %

Two findings, one against us and one for us. Against us: with a small generator, letting the model insert images while writing beats post-hoc selection on F1 (61.71 against our best 55.91). The generator places images with the full answer in view, while our selector ranks against the question alone; what our pipeline retains is precision (63.91 frozen, ten points above arm A) and marginally better prose (ROUGE-L 0.277 vs. 0.261) — consistent with figure duty and writing duty interfering when one model does both. For us: the admission-rule study's prediction replicated out of sample. Section VII's rule was fixed on the score dump before this experiment existed; applied here unchanged, it lifts arm B from 48.30 to 55.91 F1 by ending the 50 % silence, exactly the failure mode it was built to name. What the statistical half cannot measure is the axis the paper is about — where an image lands. The benchmark scores that axis with LLM-judged metrics, which we ran blind: all 400 answers (both arms, arm B in its pre-registered variant with each figure placed after the answer sentence matching its anchor context best) were interleaved under neutral shuffled keys and scored by a single judge model on the benchmark's four scales, the key unblinded only at scoring. Paired per-question differences with bootstrap intervals:

Metric (scale)	A − B	95 % CI	
Relevance (1–5)	+0.31	$[+0.15, +0.48]$	A wins
Effectiveness (1–5)	+0.24	$[+0.08, +0.40]$	A wins
Position (0–1)	−0.01	$[-0.09, +0.06]$	tie
Comprehensive (1–5)	+0.09	$[-0.04, +0.21]$	tie

The split is the finding. The generator selects better images — it chooses with its own answer in view, and relevance and effectiveness follow. But on position, the metric this paper is about, a mechanical join ties it exactly (48 questions to 49): anchor-derived placement is as good as the generator's own inline choice, while remaining traceable to a source position, and overall answer quality is indistinguishable. Placement, in other words, does not need the generator; selection is where the generator's advantage lives, which is precisely where Section VI-B showed a vision stage can substitute.

A second judge, from a different vendor. Because a single judge model is a single instrument, the identical blind manifest was re-scored by MiniMax-M3. The selection conclusions replicate in direction and magnitude: relevance +0.290 $[+0.113, +0.462]$ and effectiveness +0.199 $[+0.027, +0.376]$ to the generator, comprehensive again indistinguishable (+0.115, interval crossing zero). Position moves in our favour: −0.087 $[-0.162, -0.012]$, 56 questions to 36 — under this judge the anchor join places strictly better than the generator. The combined statement is the conservative one: under neither judge does the generator place better than the join. Per-metric inter-judge correlation is $r = 0.64$–0.79 with the second judge uniformly stricter by about 0.4 points, consistent with its 2.76×-permissive failure on the selection-κ exam being a threshold difference rather than a different ordering. Remaining caveats: text stand-ins rather than pixels for both judges, and the first judge shares a vendor with the evaluation tooling's author; blinding is the mitigation, and all per-item verdicts from both judges ship with the repo. Generator details and protocol deviations (fp8 quantisation, 4096-token window with whole-chunk truncation) are in the released runner.

VII. Admission Does Not Transfer; Ranking Does

Selection hides two decisions: how to order candidates, which the cross-encoder does, and how many to admit, which a rule does. The frozen configuration admits by an absolute floor on the sigmoid score, $s \geq 0.1$, and the external runs indict that floor from two directions: it is the difference between 11.31 and 74.03 on MMDocRAG News (Section VI-A), and it is implicated in the Lifestyle loss, where abstention caps recall on multi-gold questions. An absolute cut assumes the score distribution is corpus-invariant. It is not: description style shifts it wholesale.

The scale-free alternative is admission relative to the question's own distribution: admit candidates scoring at least $\alpha \cdot \max_j s_j$ for the question at hand. If the failure really is distribution shift, one α should transfer where no absolute floor does.

We test this without touching the model: the cross-encoder's raw scores for every candidate of every question on both benchmarks — seven evaluation units (three MRAMG domains, four MMDocRAG domains) — are persisted once, and every rule thereafter is arithmetic over that file. This also disciplines the method: the entire sensitivity surface is computed across all units at once and reported whole. Nothing is selected per benchmark, and the statistic a rule is judged on is its worst unit and its macro average — transfer, not any single win.

Table IV reports the surface, and its reading is not the one we expected when we built it. We expected relative admission to be the fix. It is a fix — but the load-bearing boundary runs elsewhere: between rules that reference an absolute score and rules that are scale-free. The frozen floor scores macro 51.45 with a worst unit of 11.31; on News it leaves the system mute on 86 % of questions, buying precision 100 at recall 6. Weakening the floor to 0.001 — three orders of magnitude — still leaves News 48 % silent, because the score distribution there sits almost entirely below even that: the shift between corpora is

TABLE IV
Admission rules over identical cross-encoder scores, F1 per evaluation unit. Every rule sees the same ranking; only how many candidates are admitted changes. Macro averages the seven units; worst is the transfer statistic. The full grid ($\alpha \in [0.2, 0.9]$, $K \in \{2, 3, 4\}$, three guard variants) is in the released artifact; rows here are representative and nothing was selected per unit.

	MMDocRAG				MRAMG				
Rule	Acad.	Fin.	News	Res.	Acad.	Life.	Web	Macro	Worst
$s \geq 0.1$, top-2 (frozen)	59.38	61.65	11.31	62.84	51.93	40.47	72.58	51.45	11.31
$s \geq 0.001$, top-2	63.32	63.40	42.64	61.97	65.55	49.26	94.97	63.02	42.64
top-2, no threshold	62.60	62.22	74.03	60.65	64.16	49.73	100.00	67.63	49.73
$s \geq 0.2 \cdot s_{\max}$, top-3	66.74	67.33	65.62	65.78	68.62	48.78	90.99	67.70	48.78

not a matter of degree an absolute constant can survive. Every scale-free rule, by contrast, lands within about one macro point of every other: plain top-2 at 67.63, relative admission at 67.70, and the entire α-grid from 0.2 to 0.9 between 61.99 and 68.06 with no unit collapsing anywhere. The choice between the two families is worth sixteen macro points; the choice within the scale-free family is worth about one.

What relative admission does buy over plain top-K is precision at equal F1: on the description-rich MMDocRAG domains it trades a few points of recall for five to eight of precision (Financial 52.65 → 60.08, Academic 63.78 → 67.54), which is the trade a system showing figures to a reader wants. It also degrades gracefully: because the cut references only the question's own score distribution, it cannot go silent wholesale the way the floor does on News.

Two disciplines kept this honest. Every number above comes from one persisted score file, so no rule enjoyed a fresh inference run; and the grid was evaluated across all seven units simultaneously, judged on macro and worst unit — a rule that wins its best unit and collapses elsewhere is exactly what this protocol exists to catch, and the frozen floor is that rule.

The frozen configuration's own numbers elsewhere in this paper are not restated under better admission: they remain as frozen, losses included. This section's claim is about rule families, established on the full surface, not a post-hoc upgrade to the headline.

VIII. Selection: Composing Sight with Discrimination

Our judge looks at the picture and agrees with a human at $\kappa = 0.552$; our selector reads a 300-character description and never sees the image. That asymmetry motivated a direct comparison, on the 48 human-annotated items — the only gold no model produced (Table V).

The gain from sight is recall, not precision: the VLM finds four of every five figures a person chose, the cross-encoder about a third. Descriptions discard enough to lose half the correct figures — but seeing the image does not make a model better at rejecting wrong ones. Composing the two therefore beats either: sight decides what is possible, discrimination decides what is best. Precision nearly doubles over the best single method and is 19× the deployed system.

TABLE V
Figure selection against human annotation. Preliminary: $n = 48$ items carrying 19 positive links, one annotator; the externally scored version of this comparison is Section VI-B.

System	P	R	F1
VLM filter → rerank → top-1	0.542	0.684	0.605
VLM selector alone	0.283	0.789	0.417
Cross-encoder alone	0.304	0.368	0.333
Deployed production system	0.028	0.211	0.049

TABLE VI
Candidate prefilter, 165 questions. N is how many of the ranked candidates the vision model is allowed to see. Latency below N = all is projected from measured per-pair cost.

N seen	4B F1	4B lat.	8B F1	8B lat.
1	0.721	0.7 s	0.735	1.0 s
2	0.736	1.3 s	0.753	2.0 s
3	0.736	2.0 s	0.757	3.0 s
all (6)	0.735	3.9 s	0.763	5.9 s

A. Showing the model fewer images

The composed pipeline costs one vision call per candidate. Because prefill scales with images rather than calls, shrinking the model is the wrong lever: a 4B model saves 3.7 s and loses half its recall. Cutting images is the right one. Letting the cross-encoder discard the obvious losers first (Table VI) costs 0.010–0.021 F1 for a two-thirds latency reduction.

Two candidates are the whole pipeline. The four the model never sees were ones it would have rejected anyway: it was spending 80 % of its time confirming the bottom of the cross-encoder's list. The prefilter's own ceiling explains where this stops being safe — the correct figure survives a cut to the top 2 for 86.8 % of links, but only 76.5 % at $N = 1$, and recall follows it down.

B. The judge prompt is the largest single gain

Replacing a permissive judging prompt with a strict one — same model, same call, different words — moves precision from 0.690 to 0.806 (F1 0.763 → 0.800) on 165 questions. This is the largest improvement in the study and costs nothing at serving time.

We attach a caveat that a reader should weigh heavily. That comparison is scored against a gold standard built by a frontier VLM, which is circular for a pipeline containing a VLM. On the 48-item human gold the strict and permissive prompts differ by 0.014 F1 against measured run-to-run noise of 0.014 — they are indistinguishable. The honest reading is that the strict prompt reliably makes our selector agree with the reference model more often; whether it makes the selector agree with a person more often is untested at this sample size.

IX. Negative Results

We report these because they were run, not because they worked.

Reranking against the generated answer instead of the question is substantially worse (F1 0.183 vs. 0.288), and concatenating the two also hurts (0.205). The likely mechanism is dilution: five sentences of tutor prose carry connective material matching almost any figure, while the question is short and discriminating.

Adding surrounding context to the figure's index text is worse than the description alone (0.248 vs. 0.270); context alone collapses to 0.119. This kills a hypothesis we had carried for some time — that the gap between MRAMG (67 %) and our textbooks (30 %) was one of text representation. Had it been, context mode would have closed it. It is the worst of the four representations.

A newer reranker is worse here. Qwen3-Reranker (2025) ranks our figure descriptions at gold-first 0.288 (MRR 0.481) against 0.788 (MRR 0.859) for the 2024 bge-reranker-v2-m3. Being newer and stronger on public multilingual benchmarks did not transfer to ranking short Indonesian figure captions.

We initially reported a far larger gap (F1 0.034 vs. 0.260) and withdrew it: that comparison applied one absolute score threshold to two models whose score distributions differ by an order of magnitude, so it measured calibration rather than ranking quality.

X. Statistical Treatment

Selection differences are tested with an exact McNemar test on per-question hit (a paired binary outcome), and precision differences with a paired bootstrap over questions (10^4 resamples). Two conventions matter for reproducibility.

Vacuity. A question on which a system emitted nothing contributes no precision term for that system. Counting silence as precision 0 would penalise abstention twice — once in recall, once in precision — and abstention is a design goal here, not a defect.

Pairing. Precision is compared only on questions where both systems emitted, so the comparison is paired rather than aggregate.

On the 165-question set the composed selector's precision advantage is significant against every comparator we ran ($p < 0.005$). We decline to report significance for the 48-item human gold: with 19 positive links, the test has too little power to distinguish the effects we care about, and reporting a p-value there would imply a resolution the data does not have.

TABLE VII
Every head-to-head comparison in the paper, including the loss.

Setting	Gold	Ours	Best other	Δ
External — gold authored by the benchmark's authors				
MRAMG Acad., IP	theirs	69.12	65.28	+3.84
forced emission	theirs	67.50	65.28	+2.22
MMDoc., gate vs. no gate, IF1	theirs	68.91	61.10	+7.81
External — where the method loses				
MRAMG Lifestyle, IP	theirs	55.00	62.23	−7.23
End-to-end IF1 vs. inline insert	theirs	55.91	61.71	−5.80
MMDoc. vs. GPT-4.1, IF1	theirs	68.91	80.7	−11.8
MMDoc. ranker vs. BM25, IF1	theirs	63.38	63.48	−0.10
Internal — gold from human annotation, $n = 48$				
Selection F1	human	0.605	0.417	+0.188
Selection precision	human	0.542	0.304	+0.238
vs. deployed system	human	0.605	0.049	+0.556
Where the method loses				
Placement rule	layout	—	MRAMG	—

Withdrawn claims. Three differences we initially reported did not survive testing and are retracted here rather than quietly dropped: a chunking gain ($p = 0.096$), anchor over co-embedding ($p = 0.670$), and a hybrid-over-anchor improvement ($p = 0.468$).

XI. Ingestion Cost

Anchoring adds no inference: it is a join over parser output. The cost of the pipeline is therefore the layout parse itself. On the partner's GB10 host, sharing the GPU with a serving language model and two embedding services, extraction runs at 1.22 s per page (192-page book in 3 min 55 s), plus 76 s to chunk, embed and store 585 chunks.

Fifteen textbooks (149 MB, Kurikulum Merdeka grade 7) ingested in 84 minutes with no failures, yielding 1,254 anchored figures where the previous pipeline had produced none.

Rendering resolution is a weak lever: dropping from 300 to 180 dpi cuts pixels to 36 % but extraction time by only 17 % (235 s → 194 s), because the cost is dominated by the number of blocks — 2,890 across 192 pages, each requiring its own recognition pass — rather than by page resolution. Increasing the KV-cache budget does nothing: the engine already reports capacity for 258 concurrent sequences against a 192-page batch.

XII. Summary of Comparisons

Table VII collects every comparison in which this method is placed against an alternative, including the one it loses. Each row names the gold standard, because the comparisons are not equally strong: the first two are scored against a gold authored by other people, the rest against our own annotation.

On MRAMG-Bench's Academic subset a 568M-parameter cross-encoder over anchor-derived context reaches Image Precision 69.12, ahead of GPT-4o at 65.28, Claude-3.5-Sonnet at 62.17 and Gemini-1.5-Pro at 59.85. Because Image Precision counts only emitted images, abstention could buy the score; ours abstains on 49 % of questions. Forcing exactly one image on every question still leads, at 67.50, with recall rising from 41.59 to 61.64. The margin is therefore not an artefact of selective answering.

On the textbook corpus, against human annotation, composing the vision-language gate with the cross-encoder reaches F1 0.605. The two stages alone reach 0.417 and 0.333. The decomposition is the finding: the vision stage supplies recall (0.789 alone, against 0.368 for the cross-encoder) and the cross-encoder supplies precision. Neither ordering of a single stage reaches the composition.

The rows the method loses. Four external comparisons go against us and are listed with the wins. Lifestyle is a genuine loss of the frozen configuration, most of it located in the admission rule (Section VII). The GPT-4.1 row compares a 568M ranker plus one gate call against a frontier LLM reasoning over interleaved candidates — we close 7.8 of the 19.6-point gap with the gate and state the rest. The BM25 row is a tie we report as a loss of the ranker: where descriptions are author-written, the cross-encoder earns nothing over lexical matching, which is precisely why the MRAMG win must be credited to the anchor representation. The end-to-end row (Section VI-C) is the newest and the most instructive: with the same small generator, inline insertion beats post-hoc selection on F1, and what our pipeline keeps is precision and provenance. Finally, anchoring alone does not beat similarity-based placement: the best placement rule in our comparison is MRAMG's bipartite assignment, not ours. The contribution of anchoring is that it makes uncaptioned figures reachable at all — a candidate-admission gain, not a ranking one — and the practical improvement then comes from selection.

XIII. Discussion and Threats to Validity

What the evidence supports. C2 and C3 are established model-free and replicated across two extraction pipelines; they do not depend on any system run. C4 divides, and both halves now carry external support. That selection carries the practical gain is corroborated independently by MRAMG, where a cross-encoder over anchor-derived context alone beats the published systems. That the gain requires a stage which looks at the image was, in the first version of this study, supported only by 48 items annotated by one person; it is now also supported externally, by the MMDocRAG composed run (Section VI-B), where the gate adds +13.65 precision on gold we did not author — with the caveat stated there that the external gate instance is a frontier VLM rather than the deployed 8B-class model. C5 is the best-evidenced claim in the paper: the rule surface was computed whole on 6,819 questions across seven domains, and its prediction then replicated on an experiment that did not exist when the rule was fixed (Section VI-C). C1 is the weakest of the five: on questions answerable only from a figure, no configuration recovered the content, and the system that most exceeded a text-only baseline did so with a significant loss of faithfulness.

The sample size is the principal threat. The headline precision of 0.542 rests on 48 items and 19 positive links. Measured run-to-run variation at that size is ± 0.014 F1, so differences smaller than that are not resolvable, and we decline to claim any. The MRAMG result carries the generalisation weight precisely because it is external, larger ($n = 200$) and authored by others.

A judge does not substitute for the annotator at this scale. We tested whether judge labels could stand in for annotation without importing the judge's bias, using a rectified estimator in the manner of prediction-powered inference [11]: the bias is measured on the human-labelled subset and subtracted, so an uninformative judge degrades the estimate to the human-only one rather than to a wrong one. A frontier VLM judged all 165 scored questions under the annotator's protocol, reusing the annotator's own eight-candidate pools verbatim for the 45 questions where one exists. Agreement was $\kappa = 0.556$ over 360 pairs, independently replicating the 0.552 recorded earlier for the same model and protocol, and the labelled pairs are representative (positive rate 4.7 % against 4.9 % across the full protocol).

The estimator applies, and it buys nothing. Four of five systems show no interval narrowing and one widens by 19 %. The cause is structural: PPI variance behaves as $\mathrm{Var}(\text{rectifier})/n + \mathrm{Var}(\text{cheap})/N$, and the second term vanishes only for $N \gg n$. At $N/n = 165/45 = 3.7$ the rectifier's own sampling error dominates. Judging every question in the corpus would reach $N/n \approx 10.8$, which is the cheap experiment that remains; annotating more items is the expensive one that still binds.

Circular gold. Our 165-question scaled gold was built by a frontier VLM. For any pipeline containing a VLM stage this inflates results — measured directly: the same selector scores 0.521 against that gold and 0.283 against human annotation. Numbers from the scaled gold are reported as consistency checks, never as effect sizes.

Single corpus, single language. All internal results are Indonesian K–12 material. The external benchmark is English. We make no claim about intermediate settings.

No online evaluation. At the time of writing, field observation consists of a handful of real queries. The 0.542 is an offline number.

XIV. Limitations

Retrieval surfaces the chunk holding the relevant figure for only 51.8 % of figure-bearing questions, which bounds every system reported here. Anchoring alone does not beat similarity-based selection; the win requires a competent placement rule, and in our comparison the best such rule is not ours but MRAMG's. 37 % of anchored figures in the deployed corpus carry a caption with no usable description — a photo credit or a synthetic placeholder — so the caption-ranked stage cannot order them by content. We identified this as a structural risk but did not demonstrate that it costs a correct figure in practice; an investigation intended to confirm it instead falsified the specific case, and we report it as an open risk rather than a finding.

XV. Conclusion

Where a figure belongs is decided at ingestion, not at ranking. Every layout parser knows it; conventional pipelines throw it away and then attempt to reconstruct it from captions that four figures in five do not have. Preserving the reading-order anchor is a join, not a model, and it is recoverable for 100 % of figures.

The external evidence divides cleanly along the paper's own axis. On placement, a mechanical anchor join ties the generator's own inline choice under blind judging while remaining traceable to a source position — placement does not need the generator, and it does not need captions. On selection, a 568M cross-encoder over anchor-derived context tops every published system on MRAMG's Academic domain (69.12 against 65.28; higher, though not statistically resolvable at $n = 200$), the anchor-window sweep confirms the mechanism by destroying it, and a stage that looks at the image adds +13.65 precision on gold we did not author. Two of the study's most useful results are constraints rather than wins: absolute admission thresholds do not transfer across corpora while any scale-free rule does — a pre-registered prediction that replicated out of sample — and where text about an image is already good, the ranker earns nothing over BM25, so the contribution is the representation, not the model. The comparisons the method loses, including an end-to-end F1 comparison against inline insertion, are in the same tables as those it wins.

Finally, we report that the vision stage ran zero times in production while every component test passed. We suspect that class of fault is common and under-reported, because a stage that fails by not running leaves nothing behind to find.

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